GyroCube | Control Settings Guide

IMPORTANT: The supplied settings have already been tuned for GyroCube. The coefficients included with the firmware correspond to a working cube configuration. No adjustment is required for normal operation. The parameters described in this guide may be changed for experimental purposes only, to observe their effect on stability and motor behavior. Inappropriate adjustments can cause oscillations, excessive wheel acceleration, or the cube to fall.

1. Operating principle

GyroCube maintains its balance using reaction wheels. When a wheel accelerates, it exerts a reaction torque on the cube. The firmware measures tilt and angular velocity, then accounts for wheel speed and accumulated wheel movement to calculate the motor commands.

These settings are often called “PID” for convenience. More precisely, the control command combines state-feedback terms: angle, angular velocity, wheel speed, and accumulated wheel movement. The K1–K4 coefficients therefore do not correspond one-to-one to the Kp, Ki, and Kd gains of a conventional PID controller.

2. Vertex control: balancing on a corner

For each tilt axis, X and Y, the firmware uses the following expression:

Command = K1 × angle + K2 × filtered angular velocity + K3 × wheel speed + K4 × accumulated movement

The result is limited to the range −255 to +255 before being distributed among the three motors. The signs of the axis commands depend on the motor-mixing calculation.

Coefficient	Measured quantity	Main effect
K1	Cube tilt angle	Restoring action related to deviation from the balance point.
K2	Filtered angular velocity	Responds to ongoing tilting; contributes to damping.
K3	Wheel speed projected onto X or Y	Limits or adjusts wheel dynamics.
K4	Accumulated wheel speed	Slow correction related to cumulative wheel movement.

3. Possible effects of changing the coefficients

Parameter	Value too low	Value too high
K1	Insufficient restoring action; greater tilt.	Abrupt reactions and possible oscillations.
K2	Insufficient damping; persistent oscillations.	Excessive response to rapid changes or sensor noise.
K3	Less pronounced wheel-speed control.	Correction may slow the wheels too much or disturb balance.
K4	Wheel drift may be less effectively compensated.	Excessive slow correction; drift or low-frequency oscillations.

These tendencies are not universal: depending on measurement signs and interactions between coefficients, a change may have different effects. Observation on the actual hardware is essential.

4. Rotation about the Z axis (Vertex)

Z command = zK2 × (Z angular velocity − rotation target) + zK3 × accumulated Z movement

Coefficient	Function
zK2	Determines how strongly the controller corrects the difference between measured rotation and the target.
zK3	Adds a correction related to accumulated wheel movement about Z.

When automatic Vertex rotation is enabled, the target may increase gradually after a period of stability. zK2 is not the target speed: it sets the response to speed error. The desired speed is determined in part by VERTEX_SPIN_SPEED, while target ramping is controlled by VERTEX_SPIN_RAMP.

5. Edge control: balancing on an edge

Command = eK1 × angle + eK2 × filtered angular velocity + eK3 × active wheel speed + eK4 × accumulated movement

The firmware uses the motor associated with the active edge. These four parameters serve functions similar to those in Vertex mode, but the geometry and dynamics are different.

Coefficient	Role
eK1	Restoring action based on tilt angle along the edge.
eK2	Response to angular tipping speed.
eK3	Accounts for the active wheel's speed.
eK4	Correction associated with that wheel's accumulated movement.

6. Initial values defined in ESP32.h

Vertex (X/Y)	Z rotation	Edge
K1 = 180.00	zK2 = 8.00	eK1 = 190.00
K2 = 30.00	zK3 = 0.30	eK2 = 31.00
K3 = 1.60		eK3 = 2.50
K4 = 0.008		eK4 = 0.014

These are the values declared in the configuration file of the firmware version examined. Coefficients stored in EEPROM may differ and can be restored by the firmware. The values actually in use should therefore be checked in the application.

7. Recommended experimental procedure

- Record and preserve all working settings before the first test.
- Change only one coefficient at a time, using a small increment.
- Perform a short, controlled test, then observe tilt, oscillations, wheel speed, and drift.
- Compare the result with the initial behavior; immediately restore the previous value if performance deteriorates.
- Test Vertex, Z rotation, and Edge settings separately. Do not automatically transfer values between modes.
- Save new values only after repeated successful validation.

Safety: Provide a way to quickly cut motor power or disable motor commands. Avoid testing near people or fragile objects. A command limited to ±255 does not guarantee that sudden movement cannot occur.

8. Key points

Stability depends on the balance between corrections for tilt angle, angular velocity, and wheel movement. A higher coefficient is not necessarily better. For normal use, retain the validated GyroCube settings; the adjustments described in this guide are intended for experimentation only.